

# Plant Nursery App — Screen-Flow Design

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**Date:** 2026-06-09 **Status:** Phase 1 flows approved · Phase 2 sketched (deferred) **Source spec:** Mobile Application for Plant Nursery.md

## Scope

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Design the screen-flows for an Android app for the Royal Botanic Gardens Melbourne volunteer nursery. **Phase 1 designed in depth; Phase 2 laid out only far enough to prove the navigation shell holds it** (the user chose option B in brainstorming).

Phase 1 processes: **Sell · Sync plant list · Export sales**. Phase 2 processes (deferred): New accession · Print label · Repot · Record death.

## Audience & accessibility (elderly volunteers)

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The volunteers using this app are **elderly**. Accessibility is a **primary design constraint across every screen** — Phase 1 and Phase 2 — not a later polish pass.

- **Big touch targets.** Minimum ~56dp tap targets; primary actions (Sell, Scan, Add to receipt, Finish & save) noticeably larger. Generous spacing so neighbouring buttons are never mis-tapped.
- **Big, legible text.** Large base font; respect the OS font-scale setting (never cap or override it). Body text no smaller than ~18sp; the numbers that matter (Total, Line total) larger still.
- **High contrast, clear layout.** Strong colour contrast (WCAG AA minimum), one primary action per screen, lots of whitespace, no clutter. Icons are always paired with a text label — never icon-only.
- **No flickering.** No flashing, blinking, or rapid-flash content (WCAG 2.3.1). Stable layouts that don't jump/reflow as data loads; a steady scanner viewfinder; calm spinners and toasts, not jarring ones.
- **Tap, don't gesture.** Every action is a visible, labelled button. Avoid swipe / long-press / pinch. Specifically **no swipe-to-delete** — removing a line is *tap line* → *explicit Remove button* (as drawn in the Sale flow). If a gesture is ever truly unavoidable, it must have an equivalent on-screen button.
- **Clear feedback & forgiveness.** Obvious pressed-state on every tap; confirm destructive actions; no auto-dismiss timeouts that rush the user.

## Decisions locked during brainstorming

| #  | Decision           | Choice                                                                                                                                                                                                                         |
|----|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Home model         | <b>Action-first</b> — home is an action grid; "Sell" opens a scan-driven receipt builder                                                                                                                                       |
| 2  | Connectivity       | <b>Offline-first</b> — selling works fully offline; <b>sales auto-export to Sheets when online</b> (plant-list update stays a manual pull)                                                                                     |
| 3  | Payment            | <b>Not captured</b> — app records the sale + total only; cash/card handled outside the app                                                                                                                                     |
| 4  | Receipt model      | <b>Receipt = one customer</b> — one "Sell" tap opens one receipt; scan every plant they buy into it                                                                                                                            |
| 5  | Discount           | <b>Per line item, a percentage</b> (changed from the source spec's flat-dollar Discount field)                                                                                                                                 |
| 6  | Unit price         | <b>Always keyed manually at sale</b> — the plant sheet carries <b>no price column</b> , so there is no pre-fill                                                                                                                |
| 7  | Not-found scan     | <b>Store the scanned barcode/accession on the line and sell as "unknown"</b> — recoverable later, never lost                                                                                                                   |
| 8  | Export destination | <b>Google Sheets only</b> — email removed from the spec                                                                                                                                                                        |
| 9  | Export/Update UX   | <b>One tap → progress → Done/Error</b> , no confirm step (manual actions only)                                                                                                                                                 |
| 10 | Auto-export        | <b>Pending sales push automatically every N min when online</b> — pre-configured, <b>default 1 min</b> , silent on success <i>and</i> failure; manual "Export now" remains                                                     |
| 11 | Receipt number     | <b>Purely local, namespaced per device</b> — each device has a configured <b>2-digit prefix</b> ; receipt # = PP-NNN (e.g. 07-241 ). Prefixes guarantee uniqueness across devices in the shared Sheet; no central coordination |

## Navigation shell

Action-first home, a persistent sync/offline status chip, and a 3-tab bottom bar.

|                                                                                                           |               |                                                              |
|-----------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------|
| Nursery                                                                                                   | ⚠ Offline · 3 | ← top bar: status chip (Synced 2h ago / Offline · N pending) |
| [ 🪴 Sell plants ]                                                                                         |               | ← primary action                                             |
| [ <span style="background-color: #007bff; color: white; padding: 2px;">NEW</span> Accession ] [ 🏷 Label ] |               | ← Phase 2 (dimmed in Phase 1)                                |
| [ 🌱 Repot ] [ 💀 Death ]                                                                                   |               |                                                              |
| ⌵ Actions   📄 Receipts   💰 Sync                                                                           |               | ← bottom nav                                                 |

## Screen inventory

- **Actions tab:** Home / action grid → each action launches its own task flow.
- **Receipts tab:** Sales history list (grouped by receipt) → Receipt detail (line items, total).
- **Sync tab:** Plant-list status + "Update plant list"; "Export to Google Sheets".
- **Global/persistent:** offline + pending-count chip in the top bar; toasts ("Saved locally", "Synced").

## Sale flow (Phase 1 — primary)

```

Home —[Sell]→ New receipt 07-241 opened (# = «device prefix»-«local seq», generated locally)
|
▼
① Scan barcode —(or)— type accession #
|
├─ found ────────────────────▶ ② Line item
└─ not found ─────────────────▶ sell as "unknown" (store scanned #, name "unknown") ▶ ② Line item

② Line item: plant name / group auto-filled
  inputs: Pots · Unit price (keyed each time — no price in sheet) · Discount %
  live Line total = pots × price × (1 - discount%)
  | [Add to receipt]
  ▼
③ Receipt 07-241 (cart): running line list + Total
  • tap a line → edit / remove
  • [+ Scan another] → back to ①
  • [Finish & save]
  ▼
④ Confirmation: receipt #, Total AUD, "Saved locally · N pending"
  [New sale] / [Done → Home]

```

### Edge paths (each a single state, not a new flow):

- **Not found:** "Not in plant list" → Retry · Type # · **Sell as unknown** (scanned barcode/accession stored on the line so it can be reconciled later).
- **Mistake / abandon:** tap a line to edit or remove; backing out of an empty receipt discards it (no save).
- **Offline:** the entire loop works; the receipt saves locally and increments the pending count.

## Sync & Export (Phase 1 — supporting)

Two "talk to the cloud" jobs. **Sales export runs automatically on a timer; the plant-list update is a manual pull.** Both manual buttons use the same pattern: **tap** → **progress (inline spinner)** → **Done or Error. No confirm, no chooser.**

Sync tab (online)

|                        |                        |
|------------------------|------------------------|
| 🌿 Plant list           |                        |
| 1,204 plants · 2h ago  |                        |
| [ Update plant list ]  | tap → ⏳ → Done / Error |
| 📄 Sales export         |                        |
| Auto every 1 min · on  |                        |
| 0 pending · last 12:04 |                        |
| [ Export now ]         | tap → ⏳ → Done / Error |

- **Update plant list (pull):** replaces the cached list from Google Sheets; shows plant count + last-updated so a scan's data is trustworthy. Stays **manual** — refreshed only when a volunteer asks.
- **Sales export (push) — automatic:** a background job pushes the **pending (not-yet-exported)** receipts to Sheets **every N minutes when online** (period pre-configured, **default 1 minute**). Export includes the transaction date (per spec). Default scope = **all pending** (no date-range picker).
- **Silent by design:** auto-export shows **no popups and no flicker** (per the accessibility rules) — the only feedback is the top-bar status chip and the tile (pending drops to 0, "last HH:MM").
- **Export now (manual fallback):** the same push on demand, keeping the one-tap → Done/Error feedback — useful before closing up or to confirm a sync visibly.
- **Done:** receipts marked exported, pending → 0.
- **Auto-export failure** (offline mid-tick, Sheets/auth): **silent** — nothing marked exported, pending unchanged, retried on the next tick. **Nothing lost.** (The visible Error screen with a "Try again" button is reserved for a manual *Export now*.)
- **Offline:** auto-export simply skips its ticks and sales keep saving locally; pending climbs and the chip shows "Offline-N". *Update plant list* / *Export now* are disabled with a plain reason ("needs Wi-Fi"). Exporting resumes automatically when back online.

**No double-counting:** exported receipts are flagged, so neither an auto-tick nor a manual *Export now* ever re-sends them.

## Phase 2 sketches (DEFERRED — revisit later)

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*Parked at the user's request; captured only to confirm the shell holds them. Not finalized.*

All four light up a dimmed home tile and reuse the same spine — **scan/pick plant** → **small action form** → **confirm** → **saved locally** → **synced later** — except New accession, which starts without a scan. Each writes a pending inventory change that the **same Sync push** sends up (matching Phase 2's auto-update-inventory goal).

- **New accession:** name / group / light / batch qty → Create & assign # → offers to print labels.
- **Print label:** scan/pick → number of labels → Add to print queue (cabled NiceLabel printer still prints).
- **Repot (tube→pot):** scan → qty repotted → Mark "Ready for Sale".
- **Record death:** scan → qty died → optional reason → Record death (subtract from inventory).

## Cross-cutting behavior

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- **Accessibility-first (elderly volunteers):** big buttons + text, high contrast, no flickering, tap-not-gesture — see *Audience & accessibility* above; applies to every screen.
- **Offline-first everywhere:** the device's cached plant list backs every scan; all changes (sales now; repot/death/etc. later) are written locally first, then pushed up — **sales auto-export on a timer**, the plant-list update is a manual pull.
- **Trust signals:** persistent status chip + pending count so volunteers always know whether data is saved-but-unsent.
- **One shared scanner + plant card** component is reused by Sell and all Phase 2 actions.

## Resolved questions (were "subject to discussion")

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All four open questions from the source spec are now answered:

1. **Unit price column in the plant sheet?** → **No**. Unit price is always keyed manually at point of sale (decision #6); there is no pre-fill.
2. **Plant-list source of truth?** → **Google Sheets**. The app pulls from and trusts the Sheet; the upstream MS Access → Sheets export pipeline is **out of scope** for the app.
3. **Receipt-number scheme?** → **Purely local** (decision #11) — generated on-device, no central coordination.
4. **Multiple devices running at once?** → **Unlikely, but assume yes**. Each device carries a configured **2-digit prefix**, so locally-numbered receipts stay collision-free in the shared Sheet (decision #11). No sync-merge logic needed — exports just append uniquely-numbered rows.